

Migration length is not a grading on Littlewood–Richardson coefficients

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Abstract

Purbhoo’s *migration* (arXiv:0705.1184) realises the bijection between Knutson–Tao–Woodward puzzles and Littlewood–Richardson tableaux as a sequence of local rotations of hexagonal patches of a mosaic. For four cycles I have treated the number of such rotations, $\ell(M)$, as a candidate q -grading on $c_{\mu\nu}^\lambda$. This note settles the question. I show that ℓ admits an exact *potential*: there is a linear height function ϕ on the $\mathbb{Z}^4$ lift of a mosaic such that every migration step increases $\phi(\diamond)$ by exactly 1. Consequently

$$\ell(M) = \sum_{\text{destination cells}} \phi - \sum_{\text{source cells}} \phi$$

depends only on the boundary data (α, β, γ) and not on M , so $G_{\lambda\mu}^\nu(q) = \sum_M q^{\ell(M)} = c_{\mu\nu}^\lambda q^{\ell_0}$ is a monomial: **the grading is trivial and the thread is dead**. A closed formula for ℓ_0 is given and verified on 361 fibres. Along the way two presuppositions of the original question are refuted: Purbhoo defines no length statistic at all (ℓ is mine, not his), and migration is a deterministic algorithm, not a rewriting system, so the confluence question (T1) as posed was ill-formed. Its well-formed replacement — does ℓ depend on the order in which the rhombi of the flock are migrated? — is answered: *no*, on all 362 mosaics computed, including 67 on which the resulting tableau *does* depend on the order. One precisely located gap remains in the proof and is stated.

1 What the source actually says

1.1 No length statistic exists in Purbhoo

The word “length” occurs in arXiv:0705.1184 only in the phrase “side length”. There is no step count, no statistic, no q -analogue and no grading anywhere in the paper. $\ell(M)$ **is my construction, not Purbhoo’s**, and the four journals that attribute “migration length” to him are wrong to do so. The only quantity he attaches to a journey is the *wake* (§4): the set of triangles and squares displaced by a travelling rhombus, a path two tiles wide. Since each rotation displaces a bounded number of tiles, ℓ is the natural normalisation of the wake, but the normalisation is mine.

1.2 Migration is deterministic

Purbhoo’s §3.1 fixes, for a given (mosaic, accessible flock F , compass direction):

- the order of the rhombi — the standard order of the flock’s tableau, which is total (equal entries lie in distinct columns);
- the hexagon — “the smallest hexagon in which $\diamond$ is contained...there is always a *unique* minimal hexagon which is entirely weakly right of the leftmost point on $\diamond$ ”;
- the rotation — 180° for the 2-fold case, and for the 3-fold case “whichever way causes some edge of the rhombus to end up exactly horizontal, or exactly vertical”.

There is no choice at any point. So the question (T1) as I had posed it for four cycles — *is the migration rewriting system confluent with constant path length?* — presupposes a nondeterminism that does not exist. ℓ is well defined for a trivial reason, and no diamond lemma is needed. The Wake Crossing Lemma, which my notes nominated as the engine of confluence, is *named and unnumbered* (confirmed) and does different work: it is what makes $\diamond \mapsto \diamond'$ order preserving, hence what makes Theorem 1 true.

The well-formed residue of (T1) is the one genuine choice in the construction: the order in which the rhombi of the flock are processed. That is answered in §3.

2 An exact model

2.1 The $\mathbb{Z}^4$ lift

Every edge of a mosaic points in one of six directions, at angles 0, 30, 60, 90, 120, 150. Purbhoo’s two families are the 0-directions $\{0, 60, 120\}$ (edges of the 0-triangles) and the 1-directions $\{30, 90, 150\}$. Write

$$u_0 = (1, 0), \quad u_{60}, u_{30}, u_{90} \text{ for the unit vectors at those angles,}$$

and encode a vertex as the integer 4-tuple $v = (a, b, c, e)$ standing for $a u_0 + b u_{60} + c u_{30} + e u_{90}$.

Lemma 1. *The map $\pi : \mathbb{Z}^4 \rightarrow \mathbb{R}^2$, $\pi(a, b, c, e) = a u_0 + b u_{60} + c u_{30} + e u_{90}$, is injective.*

Proof. $x = a + \frac{b}{2} + \frac{c\sqrt{3}}{2}$ and $y = \frac{b\sqrt{3}}{2} + \frac{e}{2} + e$. The irrational part of x determines c , that of y determines b , and the rational parts then determine a and e . $\square$

So a mosaic may be handled as an exact combinatorial object with no floating point: vertices are 4-tuples, equality is tuple equality, and comparisons of coordinates are sign tests on $p + q\sqrt{3}$ with $p, q \in \mathbb{Z}$. The tiles are

$$0\text{-triangles, } 1\text{-triangles, } \text{mixed parallelograms } \{v, v+X, v+X+Y, v+Y\},$$

X a 0-direction and Y a 1-direction; the parallelogram is a *square* when the angle is 90° — exactly the pairs (u_0, u_{90}) , (u_{60}, u_{150}) , (u_{120}, u_{30}) — and a 30/150 *rhombus* otherwise.

The hexagon $A'AB'BC'C$ of §2.4 is reconstructed by walking the boundary with exterior angles 30° at A, B, C and 90° at A', B', C' , with $E_A = u_0$, $N_A = u_{150}$; the walk closes, which is the first check on the model.

2.2 Controls

A backtracking tiler enumerates all tilings of the hexagon, subject to the nest-packing condition (all rhombi in the three nests, each a Young diagram anchored at the nest corner).

Positive control. For every boundary (α, β, γ) with $|\alpha| + |\beta| + |\gamma| = d(n - d)$ in $\text{Gr}(2, 4)$, $\text{Gr}(2, 5)$, $\text{Gr}(3, 5)$ and $\text{Gr}(3, 6)$ — 628 boundaries — the number of mosaics equals the Littlewood–Richardson number computed by two independent mechanisms: the KTW puzzle transfer matrix (`git/puzzles/lr_coefficients.py`) and a direct enumeration of LR tableaux written for this note. **628 boundaries, 0 mismatches.** The control includes a fibre with $c = 2$ and λ, μ both of length ≥ 2 , namely $\lambda = (3, 2, 1), \mu = (2, 1), \nu = (2, 1)$ in $\text{Gr}(3, 6)$. (The brief asked for a control with $c \geq 3$; no such fibre exists in any Grassmannian whose hexagon is tileable in the time available — see §6.)

Negative control. Boundaries with $|\alpha| + |\beta| + |\gamma| \neq d(n - d)$, or with a part exceeding the box, return the empty list rather than crashing or returning garbage; of the 2216 boundary triples examined in $\text{Gr}(2, 4)$, $\text{Gr}(2, 5)$, $\text{Gr}(3, 5)$, 2023 fail the degree condition, and all 2023 return the empty list.

2.3 What had to be reconstructed

Two points of §3.1 could not be implemented from the text alone, and I record them as deviations rather than hiding them.

Remark 2 (the 3-fold tie-break). Purbhoo’s rule — rotate “in whichever way causes some edge of the rhombus to end up exactly horizontal, or exactly vertical” — cannot by itself select the forward rotation. Concretely, in a traced journey in $\text{Gr}(3, 5)$ two consecutive steps require incompatible frames: the first forces the frame angle $\theta \in \{60^\circ, 150^\circ\}$ and the second forces $\theta \in \{0^\circ, 90^\circ\}$. So no fixed frame makes the literal rule agree with forward motion. I therefore take as primary the property Purbhoo states as the *consequence* of his rule — “the rhombus moves roughly to the right (although sometimes it moves strictly vertically)” — and use the horizontal/vertical condition as the tie-break.

Remark 3 (“unique minimal hexagon”). The minimal hexagon is not unique in the naive sense: among the 13,541 steps computed, 1,593 had two or more candidate hexagons of minimal tile count. Uniqueness is restored by Purbhoo’s own §4: the wake is a *path* from the initial to the final position of $\diamond$, so a position is never revisited inside one journey. With that filter, and the two above, **every one of the 13,541 steps had a unique legal continuation**: 9,889 were forced outright and 932 were decided by the horizontal/vertical rule. Zero ambiguous steps.

3 (T1): ℓ does not depend on the order

Fix a mosaic M . Call an ordering of the rhombi of α *admissible* if each rhombus in turn admits a legal journey to nest B .

Proposition 4. *On all 362 mosaics of $\text{Gr}(2, 4)$, $\text{Gr}(2, 5)$, $\text{Gr}(3, 5)$, $\text{Gr}(3, 6)$, the total number of rotations ℓ is the same for every admissible order. The number of admissible orders per mosaic ranges from 1 to 42; 144 of the mosaics have more than one. On 67 of them the resulting tableau does depend on the order.*

This is the sharp form of the statement: migration is *not* confluent as a map — different orders produce different LR tableaux — and yet the step count is an invariant. That dissociation is the clue, and §4 explains it.

4 The potential

Definition 5. For a vertex $v = (a, b, c, e) \in \mathbb{Z}^4$ set $\phi(v) = b + e$, and for a tile t with vertices $v_1, \dots, v_k$ set $\phi(t) = \frac{1}{k} \sum_i \phi(v_i)$.

ϕ is linear on $\mathbb{Z}^4$ but does *not* factor through π : there is no linear form $\alpha x + \beta y$ on the plane with $\phi = \alpha x + \beta y$ (matching the coefficient of a forces $\alpha = 0$, matching b forces $\beta = 2/\sqrt{3}$, and then the coefficient of c is $1/\sqrt{3} \neq 0$). So ϕ is a genuine *height* on the lifted surface, not a planar functional — which is exactly what one wants of a volume-type potential.

Theorem 6. *Every migration step increases $\phi(\diamond)$ by exactly 1: $\phi(\diamond') - \phi(\diamond) = 1$.*

Status. Verified on all 707 migration steps arising from the $\text{Gr}(2, 4)$, $\text{Gr}(2, 5)$ and $\text{Gr}(3, 5)$ sweeps, with zero exceptions, and reduced by exhaustive classification to a single unproved configuration (Gap 8). The classification is the following.

Lemma 7 (classification of steps). *Enumerate every hexagon that can occur, up to translation: the 2-fold ones are the zonogons on three of the six directions, the 3-fold ones have boundary p, q, Rp, Rq, R^2p, R^2q with R the rotation by 120° . There are 90 such regions. Enumerate every tiling of each, every rhombus in it, and every admissible rotation, subject to*

- (C1) *the hexagon lies entirely weakly right of the leftmost point of $\diamond$ (Purbhoo §3.1), hence their leftmost points coincide;*
- (C2) *the hexagon contains exactly one rhombus — during a migration all other rhombi are packed in nests;*
- (C3) *$\diamond$ carries one of the two orientations permitted by Purbhoo’s Proposition 3.1 (“the rhombi can only ever be in two of three orientations during the migration process”). For $A \rightarrow B$ these are the nest- A type, edges $\{0^\circ, 150^\circ\}$, and the nest- B type, edges $\{60^\circ, 30^\circ\}$.*

Then among all weakly-forward rotations, $\Delta\phi = 1$ in every case except one: the strictly vertical 180° step, which exists in two directions, one with $\Delta\phi = +1$ and one with $\Delta\phi = -1$.

Condition (C3) is not an extra assumption: it is a theorem of Purbhoo’s, and the computation confirms it independently — across 707 steps the travelling rhombus took exactly the two predicted orientations and no other, with the 180° steps preserving the type ($244 + 153 = 397$ of them) and the 120° steps exchanging the two ($281 + 29 = 310$). Without (C1)–(C3) the classification admits 210 forward configurations with $\Delta\phi \in \{0, \pm\frac{1}{2}\}$; with them, exactly one survives.

Gap 8. The strictly vertical 180° step always goes in the ϕ -increasing direction. Of the 707 steps computed, 91 were strictly vertical, and all 91 had $\Delta\phi = +1$; the $\Delta\phi = -1$ vertical step never occurred. I do not have a proof that it cannot. The missing ingredient is an argument that the vertical move is oriented by the global geometry (the rhombus is travelling towards nest B), which is precisely the content of Purbhoo’s Proposition 3.1 for the horizontal component and is not supplied there for the vertical one. *This is the only gap, it is finite, and it is one configuration.*

Corollary 9. *Granting Theorem 6, $\ell(M)$ depends only on the boundary data.*

Proof. A hexagon rotation displaces triangles and squares and exactly one rhombus (Purbhoo §4: the wake consists of triangles and squares). Hence the journeys of the rhombi of α are independent as far as counting is concerned, and

$$\ell(M) = \sum_{\diamond \in \alpha} (\phi(\diamond') - \phi(\diamond)) = \sum_{\diamond \in \alpha} \phi(\diamond') - \sum_{\diamond \in \alpha} \phi(\diamond).$$

The second sum is the sum of ϕ over the cells of α in nest A . By Corollary 3.2 the first sum is over the cells of λ/μ in nest B , and λ, μ are determined by γ, β . Both sums are therefore functions of (α, β, γ) alone. Note that the sum is over the *sets* of source and destination cells, not over the bijection between them; this is exactly why ℓ is insensitive to the order (Proposition 4) while the tableau is not. $\square$

Theorem 10 (closed form). *With α the nest- A partition, $\mu = \beta$, and λ determined by $\lambda^\vee = \gamma$, i.e. $\lambda_k = (n - d) - \gamma_{d+1-k}$,*

$$\ell_0(\alpha, \beta, \gamma) = \sum_{i=0}^{d-1} \left[(\lambda_{i+1} - \mu_{i+1}) \left(n - i - \frac{1}{2} \right) - \frac{1}{2} \alpha_{i+1}^2 \right].$$

Proof. On a nest- A cell (i, j) the four vertices have ϕ -values $j, j, j+1, j+1$, so $\phi = j + \frac{1}{2}$; on a nest- B cell (i, j) they are $n-i, n-i-1, n-i-1, n-i$, so $\phi = n - i - \frac{1}{2}$. Substituting into Corollary 9 and summing $\sum_{j < \alpha_{i+1}} (j + \frac{1}{2}) = \frac{1}{2} \alpha_{i+1}^2$ gives the formula. $\square$

Verification. The closed form agrees with the simulated ℓ on all 487 computed journeys and on all 361 fibres — a second mechanism (arithmetic in the boundary data) against the first (geometric simulation of the rotations).

5 (T2): the grading

Corollary 11. $G_{\lambda\mu}^\nu(q) = \sum_M q^{\ell(M)} = c_{\mu\nu}^\lambda q^{\ell_0}$. *The grading is a monomial; it carries no information beyond the coefficient itself.*

Data. Over the 361 nonempty fibres of $\text{Gr}(2, 4)$, $\text{Gr}(2, 5)$, $\text{Gr}(3, 5)$, $\text{Gr}(3, 6)$: the number of distinct values of ℓ per fibre is 1 in 361 cases out of 361.

Why that number is nearly uninformative on its own. 360 of those 361 fibres have $c = 1$, and on a singleton fibre “ ℓ is constant” is vacuously true. A sweep dominated by multiplicity-one fibres cannot see a grading at all; its all-green is of rank one. The fibres that can see it were therefore targeted separately:

$$\begin{array}{ll} \text{Gr}(3, 6) : & 1 \text{ fibre with } c = 2, \quad \ell = (11, 11) \\ \text{Gr}(3, 7) : & 10 \text{ fibres with } c = 2, \quad \ell \text{ constant in all 10} \\ \text{Gr}(4, 7) : & 10 \text{ fibres with } c = 2, \quad \ell \text{ constant in all 10} \end{array}$$

21 genuinely multiplicity- ≥ 2 fibres, all constant. Items (b) and (c) of the brief — comparison with LLT $c_{\lambda\mu}^\nu(q)$ and with the Schilling–Shimozono–White cospin generating function, and symmetry/unimodality — are not reached: a monomial matches no nonconstant family, and there is nothing to test for unimodality.

6 Limits of the computation

- The sweep covers $d(n-d) \leq 9$ plus targeted fibres at $d(n-d) = 12$. The brief asked for all $|\nu| \leq 6$; shapes needing a box larger than 3×4 or 4×3 are not covered, because the tiler is exponential in the hexagon area.
- **No fibre with $c \geq 3$ was tested.** The smallest Grassmannian carrying one is beyond the tiler. This is the brief's requested control and it is *not* discharged; what is discharged is $c \geq 2$, 21 times.
- Theorem 6 rests on Gap 8.
- Remarks 2 and 3 are deviations from the literal text of §3.1. They are justified by other statements in the same paper, and the resulting map reproduces the LR counts exactly, but they are reconstructions.

7 What this kills, and what it leaves

The four-cycle thread “ $\ell(M)$ as a q -grading on LR coefficients” is dead. It was dead for a structural reason that was visible from the start and that I never looked for: migration is a monotone process, monotone processes have potentials, and a statistic that is a potential difference cannot distinguish the objects in a fibre. The Gustafsson–Westerlund hazard named in the brief — a bijectivization weight whose bijection exists only where the grading is trivial — is the same shape, and I recorded it as the named risk before computing. It was the right risk.

Two things survive. First, the apparatus: an exact, controlled implementation of mosaics, tilings and migration, validated against the LR numbers on 628 boundaries. Second, a sharper question. The dissociation in Proposition 4 — the step count is order-independent while the output tableau is not — is a statement about Purbhoo's §5.2 open question on the monodromy of the groupoid of mosaics with flocks. ϕ is a function on that groupoid's arrows that is additive and depends only on endpoints; it is therefore a candidate obstruction class, and it is *trivial*, which is a (negative) answer to a weak form of his question. Whether there is a finer invariant of a migration path that is *not* a potential difference — and such an invariant is exactly what a nontrivial grading would require — is the question that replaces the dead one.